

Up-Counter Operation in PLC

Pages 5–7 (Lecture Notes)

1. Up-Counter Overview

An up-counter increments its accumulated value by one for each valid input transition and generates an output when a predefined number of events has occurred.

- Counting occurs on a **false-to-true (OFF to ON)** transition.
- Continuous ON signals do not cause multiple counts.
- The counter triggers an action when the accumulated value equals the preset value.

2. Practical Up-Counter Example

System Description

- PLC: Allen-Bradley SLC 500
- Input device: Pushbutton (PB1)
- Output devices: Red and Green pilot lights
- Preset value of counter: 7

Counting Operation

Each press of pushbutton PB1 produces an OFF-to-ON transition, increasing the counter's accumulated value by one.

- Accumulated value starts from zero.
- Each valid input pulse increments the count.
- Holding the pushbutton does not increase the count.

Counter Completion

When the accumulated value becomes equal to the preset value, the counter DONE (DN) bit is energized.

- Green pilot light remains ON while accumulated value $<$ preset.
- Red pilot light turns ON when accumulated value $=$ preset.

3. Ladder Logic Operation

The ladder logic diagram demonstrates how counter status bits are used to control output devices.

- Pushbutton input increments the up-counter.
- Counter DONE (DN) bit is used in subsequent rungs.
- Output states depend on the counter status, not directly on the input.

Operational Logic

- Before preset is reached: normal operation continues.
- After preset is reached: DN bit changes state and controls outputs.

4. Counter Reset Operation

A separate reset pushbutton (PB2) is used to clear the counter and restart the counting process.

- Pressing PB2 makes the reset rung TRUE.
- Accumulated count is reset to zero.
- Counter DONE bit becomes FALSE.

Counting resumes only after the reset pushbutton is released and the reset rung becomes FALSE.

5. SLC 500 Counter File Structure

In Allen-Bradley SLC 500 PLCs, counters are stored in a dedicated counter file.

Counter File

- Counter file number: **File 5 (C5)**
- Counter address range: **C5:0 to C5:255**
- Maximum counters per file: 256

Counter Element Structure

Each counter element consists of **three 16-bit words**:

Word	Purpose
Control Word	Stores status bits (DN, CU, CD, etc.)
Preset Word	Stores preset value
Accumulated Word	Stores current count value

These three words together form a single counter element.

6. Key Exam Points

- Up-counters increment on OFF-to-ON transitions.
- DONE bit becomes true when accumulated equals preset.
- Outputs are controlled using counter status bits.
- Reset clears accumulated value and allows reuse.
- SLC 500 counters are stored in file 5 and use three data words.